20 Questions with LLMs

CSE-519 Project Proposal

1 Abstract

The game of 20 questions involves a subject (person) having an object, profession or concept in their mind while players (guessers) question the subject to narrow down and guess the object, having the limit of 20 questions to do so. Our project aims to evaluate the performance of Large Language Models (LLMs) for both the subject and the player.

Throughout the years, LLMs have gotten better at emulating human conversations and at playing games at levels comparable (or even exceeding) professions. We aim to see how well they have really gotten and which LLMs in particular perform best at such tasks. Not only will we investigate their performance at the game, but we will also judge the models on how human-like they sound when posing questions. We will also look into reasons why certain LLMs are better than others, investigating how they were trained and for which original purpose they were built.

2 Background

According to C. R. S. Kumar in his paper *Large Language Models (LLM) for Automating 20 Questions Game*, the game used to be accomplished by "using decision trees, heuristic algorithms, or rule-based systems that map out potential objects and the corresponding questions that might lead to their identification. However, these methods often struggle with the complexity and variability inherent in human language, especially when dealing with ambiguous or less common objects." He goes on to utilize 1 LLM as a guesser, and another LLM as the answerer. This paper then uses the data gathered from a Kaggle competition, along with other sources.

In 2024, Kaggle hosted a competition to see who could create the best LLM for the 20 Questions game. Each team had the ability to submit bots who would play games with other bots, and the winner of the competition utilized a dual LLM approach to handle different types of questions, with a Meta Llama 3 8B Instruct as the general purpose LLM, and a DeepSeek Math model to handle complex mathematical oriented questions about the keyword. The competition featured a 2 vs 2 cooperative game, with each team consisting of a questioner and answerer. Our work will be simplifying this to a one versus one conventional implementation.

While previous work has demonstrated that LLMs can participate in 20 Questions, there is still an opportunity to investigate how different language models compare when assigned different roles.

An important consideration is that the questioner and answerer require different capabilities. The questioner must demonstrate strategic reasoning, effective question selection, and the ability to interpret previous responses. The answerer must demonstrate factual knowledge, consistency, and the ability to interpret questions accurately. Because of this, a model that performs well as a questioner may not perform well as an answerer. Since our project requires each model to perform both roles, the models chosen need to be well balanced for both of these, and the winning model may not be the best guesser, or the best answerer.

3 Approach

Our approach involves finding LLMs and researching what they were originally created for. Playing 20 questions directly with each LLM (as a player/questioner) and evaluating performance based on how many questions it takes for them to guess our object, and if they guessed correctly or not.

We will develop a scoring algorithm to score each LLM on its performance.

We will also investigate which topics each LLM performs best on.

We would like to evaluate the humanness of the questions the models come up with for which we will also have to develop a metric. This might be subjective, however, we will try our best to find characteristics that seem more bot-like, rather than natural.

Pairwise comparison: We will put pairs of LLMs head to head (even with themselves) and see how well they do

against each other and what concepts they come up with. Are the concepts repetitive?, which LLM does best when it comes to coming up with concepts more randomly?

For the questions asked, we will evaluate and score them based on how good the question was at narrowing the search down.

For each pair we will initially have a hypothesis as to how it will perform based on how it was trained. Then we will see if our hypothesis holds true, and what could explain their actual behaviour.

We will test multiple models and run brief experiments involving runs with a small set of topics, then we will identify which ones work best and use them for further analysis.

4 Models and Data

For many of the models we looked at, we were searching for four main things:

Knowledge:
How good it is with facts and scientific know-how.

Humanness:
How naturally it poses questions, as opposed to sounding like a bot.

Size: We want to investigate how well small models perform compared to larger models. Does Occom's Razor hold? Do simpler/lightweight models do the job just as well?

Logic and Reasoning:
How well it does for logic and deduction, this would be key when it comes to narrowing down the questions to reach the answer faster.

Candidate Models

Model	Characteristics	Expected relevance
DeepSeek R1 0528	Strong reasoning and inference; performs well on several benchmarks, including GPQA Diamond and major sciences.	May excel at games requiring scientific facts and deduction.
Kimi K2	Originally intended for coding; also supports conversation and step-by-step logical reasoning.	Its thinking style may be interesting to evaluate in 20 Questions.
Xwin-LM	Based on Llama 2 and optimized for human-like conversation; exceeded GPT-4 in the AlpacaEval instruction-following benchmark.	May generate more natural questions.
Command-R	Optimized for long-context conversation and grounded generation from supplied texts or documents.	May be a strong guesser when interpreting many small questions.
Cipher	Designed for consistent, human-like conversations; gives direct, precise, well-structured answers and uses low memory.	May stay focused on the topic and narrow the game effectively.
Spark-X2.5-4B	General-purpose model specialized in writing, reasoning, and conversation; has lower overhead than models with larger contexts.	Its reasoning specialization may help it perform well in 20 Questions.
Thoxhuman-chat-8b	Digital-human conversation model with adaptable personas, natural dialogue, and verified factual responses; based on Qwen3.	May guess quickly and pose natural questions.
Apollo-Astralis-4B	Built for deep logical thinking and human–AI collaboration; conversational, empathetic, and runs with as little as 4 GB RAM and 2.5 GB disk space.	May perform well through logical thinking while testing lightweight models.
Billy	"Better Interaction Language Learning Yield" model customized for casual, human-like conversation; relatively lightweight at 4.1 GB.	May pose questions more naturally.

Table 1: Candidate LLMs and the properties motivating their inclusion.

We may also use historical question sets available for 20 Questions games and see how well each model performs with those legacy questions. This would give us a good benchmark to compare each model.

There was a *Twenty Questions* show that aired in 1949 on WOR-TV and NBC. Although not all episodes are available (some records were lost) we think it would be possible to use questions and topics from this show to create our own

benchmark to test models against.

We also found the CLiPS 20Q dataset from Hugging Face, which we could use to make a similar benchmark.

5 Evaluation Plan

Our evaluation will focus on four dimensions: overall game performance, word selection, question quality, and answer accuracy. Game performance will be measured by win rate and the average number of questions needed to identify the hidden word. We will test all pairings of at least four LLMs, with each model serving as both questioner and answerer, to identify differences between roles and self-play. Word selection will be evaluated by comparing the distribution and diversity of model-selected words across categories (nouns, places, people, and general vocabulary) against a uniform random baseline. Question quality will be measured by how effectively questions reduce the remaining candidate space, using information gain to assess how closely questions divide possibilities in half. Answer accuracy will be evaluated by comparing responses against known facts about the hidden word, measuring truthfulness and consistency while identifying misleading or contradictory answers.

We also plan to identify what characteristics make sentences/questions sound more human. We will use this to come up with a metric that would help us evaluate the humanness of each LLM.

Scoring Functions

Generally we will have to develop the following scoring functions:

- Scoring functions for guess performance.
- Scoring function for how well the questions it asked were.
- Scoring on how varied the answers are.
- Scoring on how truthful the answerer is.
- Scoring on "humanness" of questions asked.
- Scoring on computational efficiency.

If time and resources permit, we will also evaluate computational efficiency using metrics such as peak RAM and VRAM usage, response time, and token consumption. This will allow us to examine the tradeoff between game performance and resource requirements, identifying models that achieve comparable results with fewer computational resources.

6 Next Steps

Currently, we have a rudimentary 20 questions game set up with 1 model playing both the guesser and the answerer. As our next steps, we should allow for multiple models to be loaded, and for multiple games to be run in sequence to allow for better data collection. After collecting the data for multiple models, and deciding on at least 4 to move on, we need to collect data for each possible combination, and score each model based on the statistics we chose earlier. If we have time later, we can try to run a larger model and see how it will compare to the regular sized models we can run on our own hardware by using Colab, or another available server. Additionally we could try to use an API model to compare how the open-source vs proprietary models can tackle this game.

Our code is located at `https://github.com/n-man13/CSE519-Final-Project`.

7 References

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